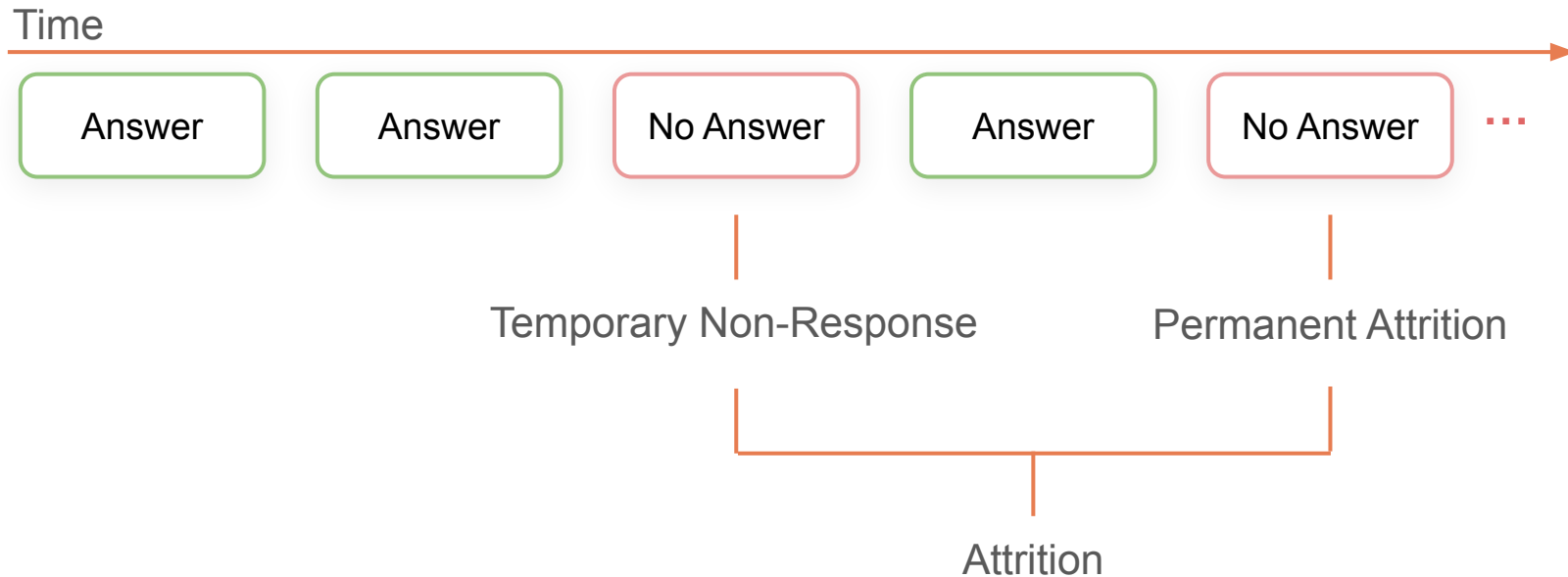


Master Seminar

Health Determinants of Panel Attrition: A SOEP Logistic Regression Analysis

By Anton Cronet

1. Introduction: What is Panel Attrition?



1. Introduction: Attrition in the SOEP

Consequences For Researchers

Attrition is a loss of data

Attrition is systematic

→ Bias enters the sample

Finding factors causing Attrition
can help with mitigation measures

Consequences For Survey

New samples are needed to keep
the Panel representative.

It is cheaper to keep participants
than finding new ones.

1. Introduction: Insight from Similar research

Grip Strength is a good indicator for physical health and morbidity in German SOEP (Steiber, 2016)

Stress is a major factor predicting panel attrition (Rübsamen et al. 2017)

Documentation of Sample Sizes and Panel Attrition in the SOEP (Kroh et al., 2018)

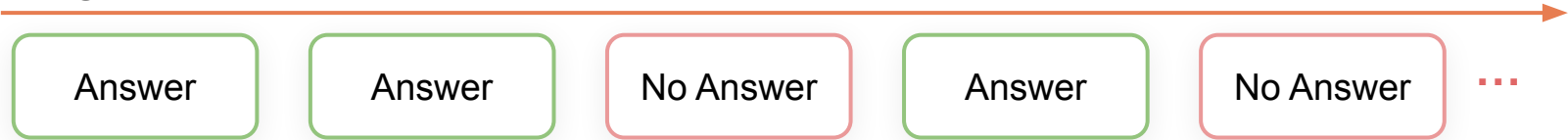
2. Research Questions

To what extent do health factors influence the likelihood of sample attrition in the German Socio-Economic Panel?

What are the implications for the SOEP study design?

3. Data: Defining Attrition

Time



Building the Attrition Variable:

- Step 1: Note the final observed Year of an individual
- Step 2: If last year is before 2023 = Attrition
(Temporary non-response $\neq$ Attrition)
- Step 3: Remove Attrition caused by death

3. Data: SF-12 Health Variables

Q1 General health

→ General Health

Q2 Moderate activities

→ Physical Functioning

Q3 Climbing stairs

→ Physical Functioning

Q4 Physical health limits work

→ Role Physical

Q5 Pain interferes with work

→ Bodily Pain

Q6 Felt calm & peaceful

→ Mental Health

Q7 Had a lot of energy

→ Vitality

Q8 Physical/emotional health limits social activities

→ Social Functioning

Q9 Emotional problems limit work

→ Role Emotional

Q10 Felt downhearted & depressed

→ Mental Health

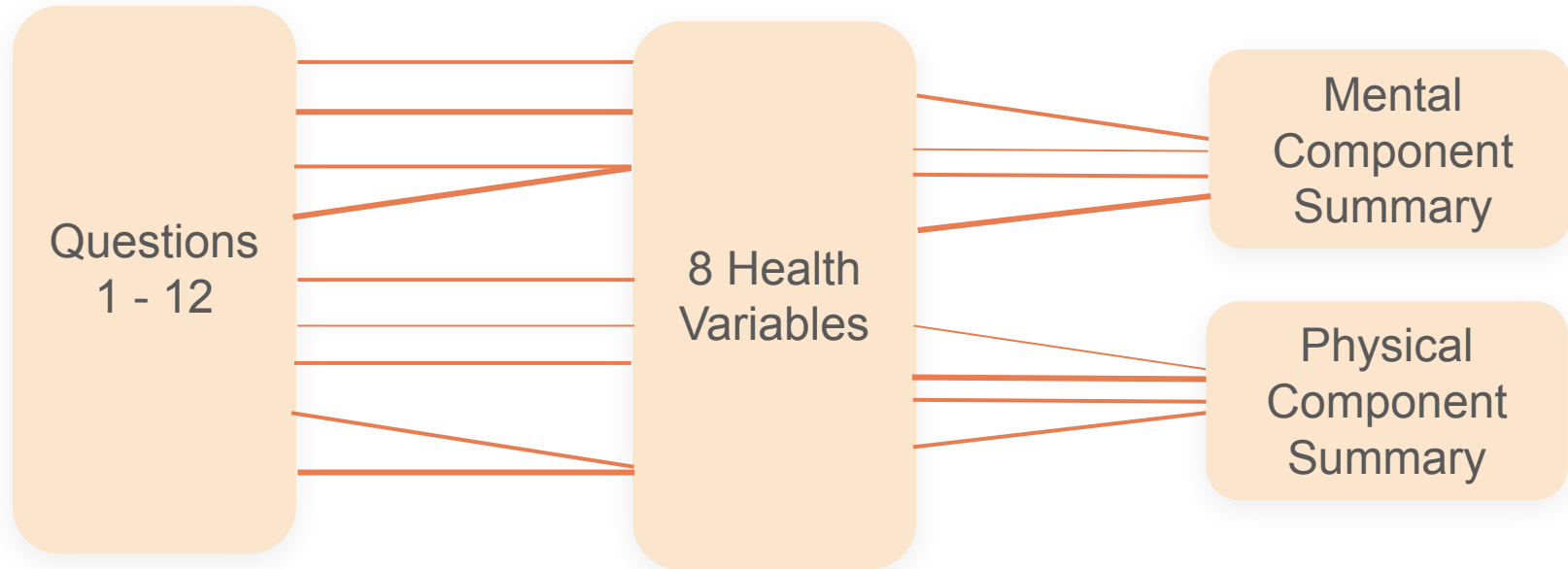
Q11 Accomplished less due to physical health

→ Role Physical

Q12 Accomplished less due to emotional problems

→ Role Emotional

3. Data: SF-12 Health Variables



3. Data: Control Variables

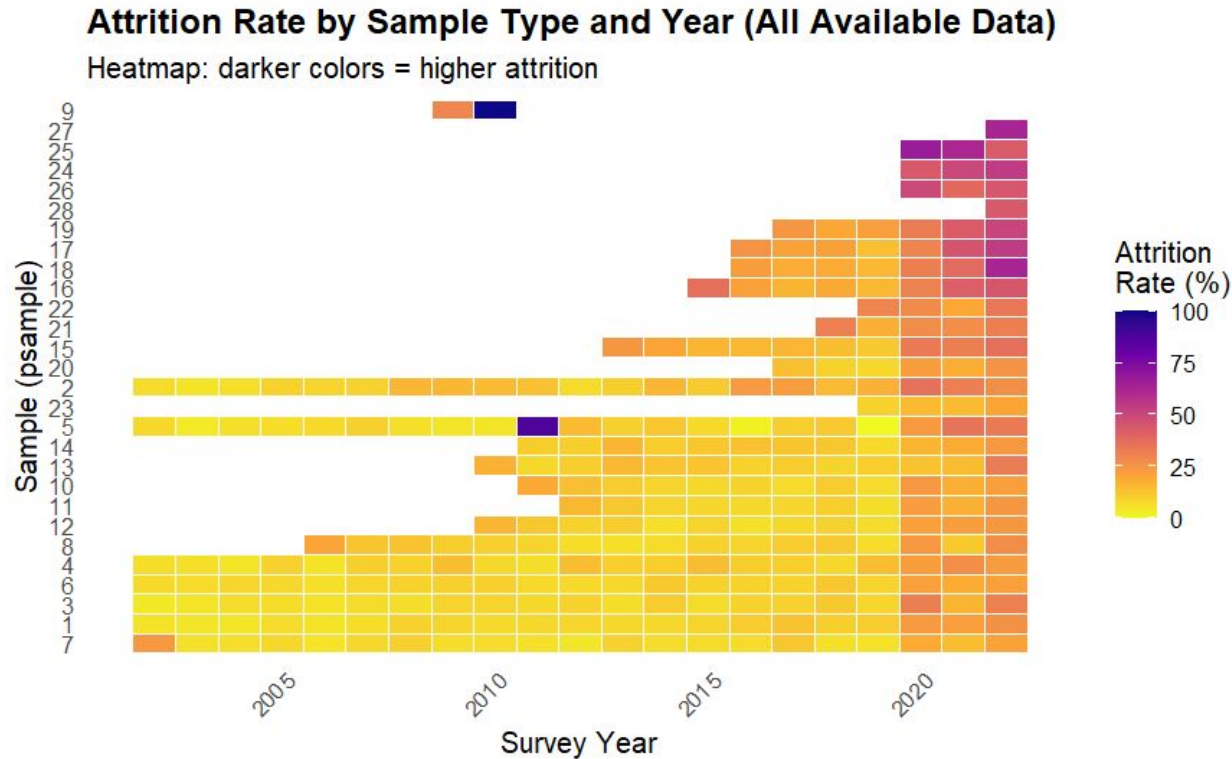
Socioeconomic Status Controls:

- Employment status
- Migration background
- Partner status
- Years of education
- Actual Work Hours / Week

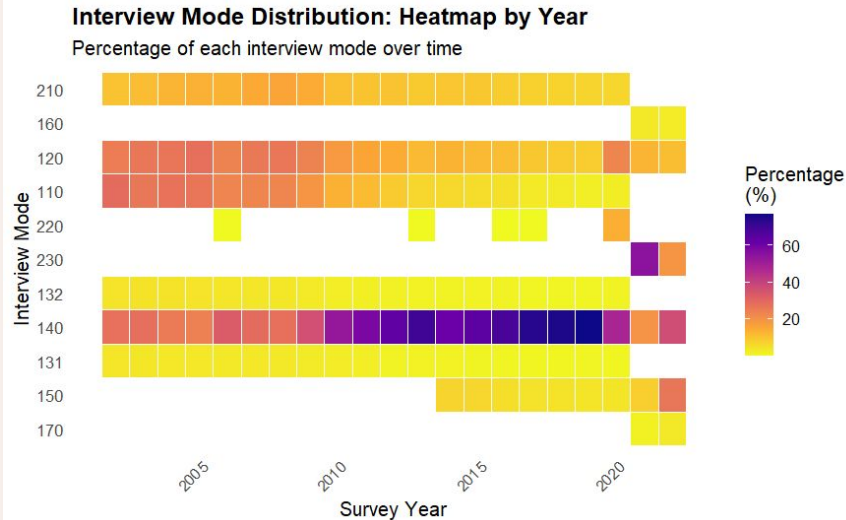
Fixed Effects:

- Survey mode fixed effects
- Sample fixed effects
- Year fixed effects

3. Data Overview: Attrition Heatmap



3. Data Overview: Interview Type Heatmap



[210] Mail: No Interviewer

[160] Telephone: Interviewer

[120] Written: No Interviewer

[110] Oral Interview: Interviewer

[220] Telephone: Interviewer

[230] Telephone: Interviewer

[132] Oral And Written: Interviewer

[140] Face to face: Interviewer

[131] Written: with Interviewer

[150] Online at home: No Interviewer

[170] Computer: with Interviewer

4. Methodology

Step 1: Identify which variables to use by comparing logistic regressions

Step 2: Recreate best fitting model using Bayesian Logistic Regression

Step 3: Analyse (Health) factors' effects on Attrition

5. Results: Five Nested Models

Model	Health Measures	Physical Function	Controls	N
1. Basic	MCS + PCS	BMI	Age, Gender	236 156
2. Subscales	8 Subscales	BMI	Age, Gender	236 156
3. + Grip Strength	MCS + PCS	BMI + Grip	Age, Gender	55 725
4. Full (subscales)	8 Subscales	BMI + Grip	Age, Gender + SES + FE	29 693
5. Full (summary)	MCS + PCS	BMI + Grip	Age, Gender + SES + FE	29 693

5. Results: Frequentist Models Key Takeaways

Model 1 vs. 2: Composite vs. Individual Health Scales

- Composite scores obscure important dimensional differences

Model 1 vs. 3: Adding Grip Strength

- Other coefficients unchanged, Grip Strength captures new information

Model 3 vs. 4: Full Specification with Controls

- Controls generate much better model fit

Model 4 vs. 5: Detailed vs. Composite Health

- 8-variable scale is superior - uncovers mental health effect

5. Results: Bayesian Model

Model Information:

Chains: $4 \times 2,000$ iterations (1,000 warmup)

Weakly informative priors

→ $N(0, 2.5)$ coefficients; $N(0, 5)$ intercept

Model Fit:

R values: < 1.02 (good chain convergence)

Pareto-k Diagnostic: < 0.7 (Reliability of LOO approximation for each observation)

Coefficient differences Bayesian vs Frequentist:

Maximum difference in coefficients:

→ 0.0037

Average coefficient difference:

→ 0.00053

SF-12 Marginal Effects

Top 3 Most Impactful Health Dimensions:

1. Mental Health:

- - 8.3% in attrition per SD of improvement
(95% CI: -13.5 to -2.9)

2. Social Functioning:

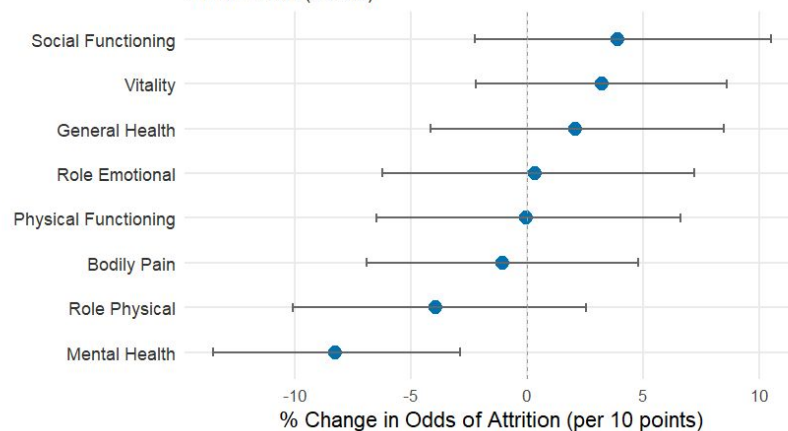
- + 3.9% in attrition per SD of improvement
(95% CI: -2.3 to 10.5)

3. Role Physical:

- - 3.9% in attrition per SD of improvement
(95% CI: -10.1 to 2.6)

SF-12 Health Subscales: Effect on Attrition

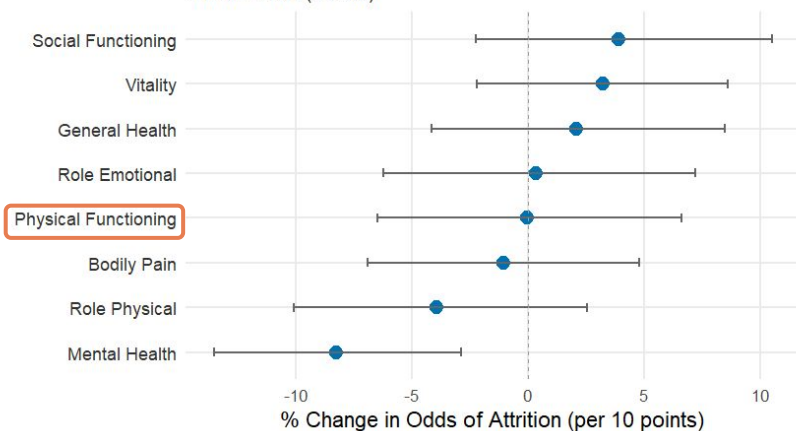
Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Health Results: Physical Functioning

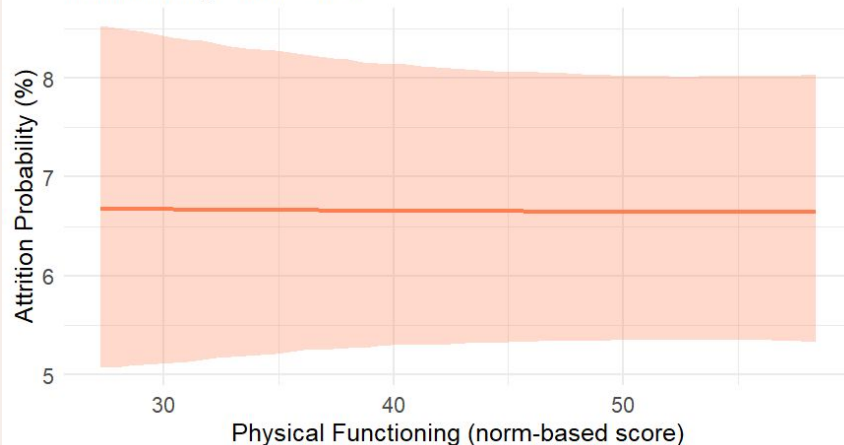
SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Physical Functioning

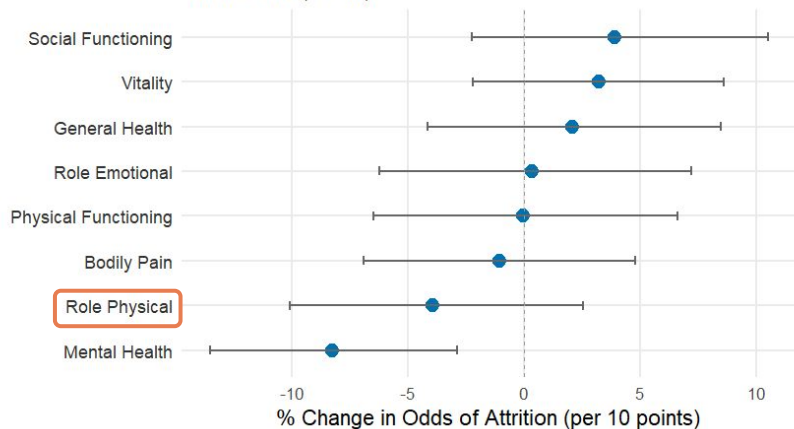
Prediction range: 6.6% - 6.7%



Health Results: Role Physical

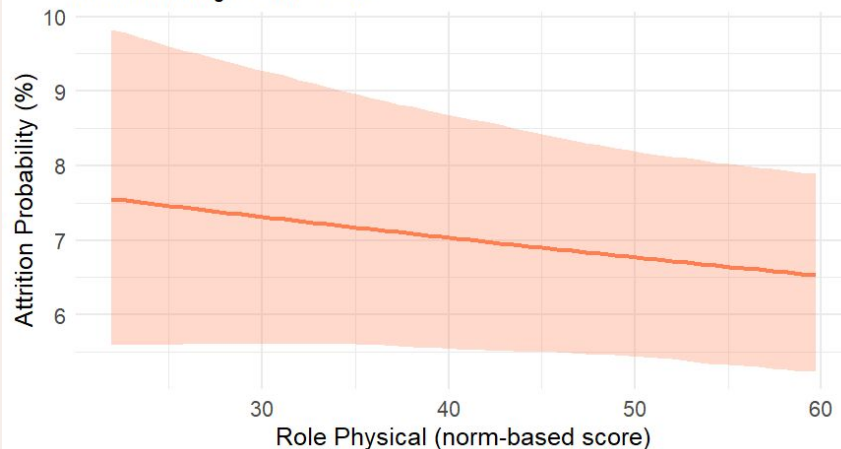
SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Role Physical

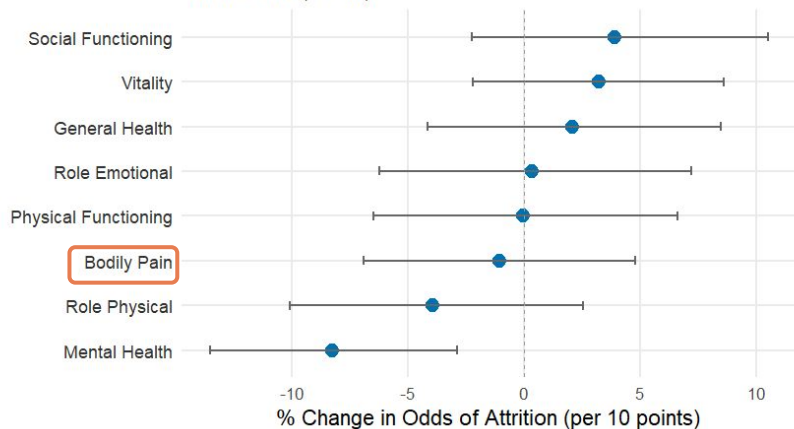
Prediction range: 6.5% - 7.6%



Health Results: Bodily Pain

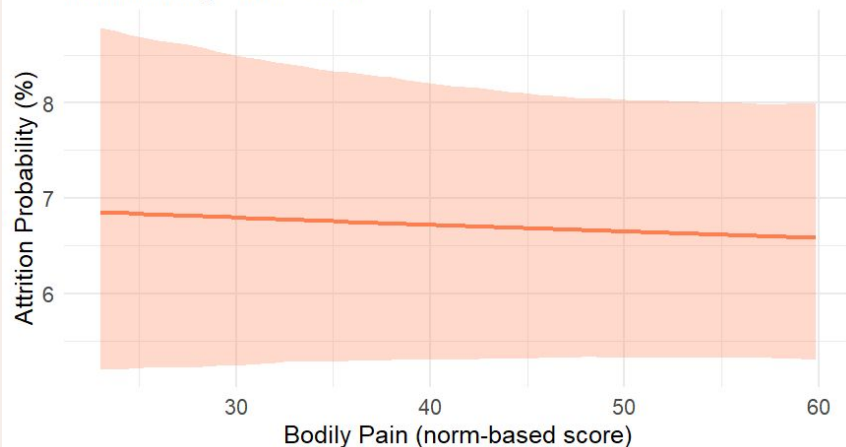
SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Bodily Pain

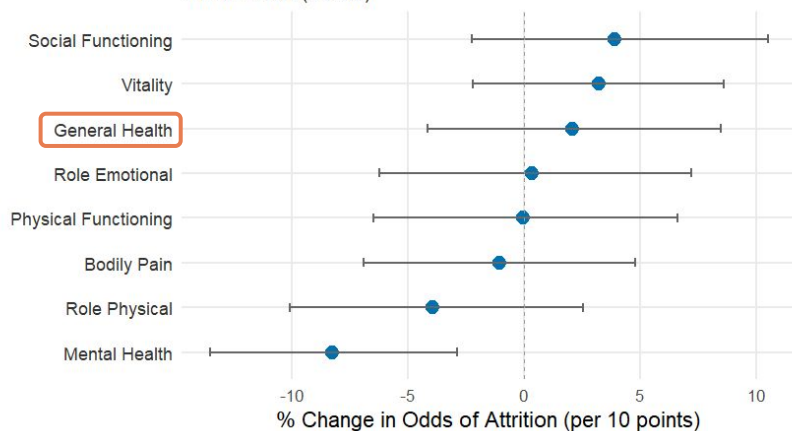
Prediction range: 6.6% - 6.9%



Health Results: General Health

SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by General Health

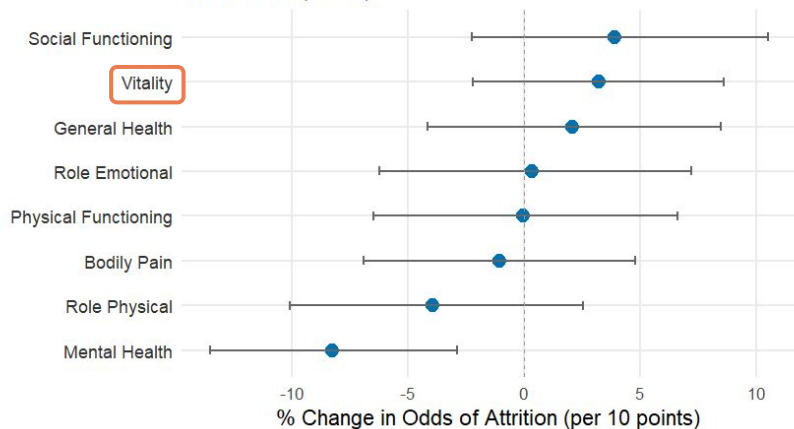
Prediction range: 6.3% - 6.8%



Health Results: Vitality

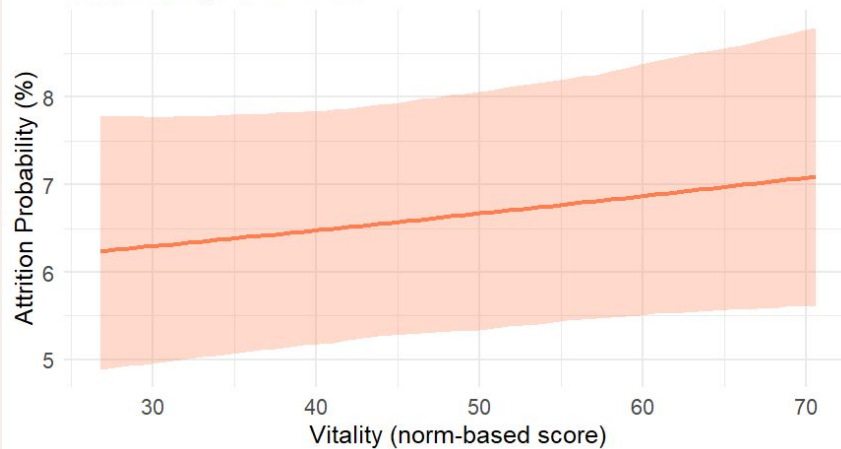
SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Vitality

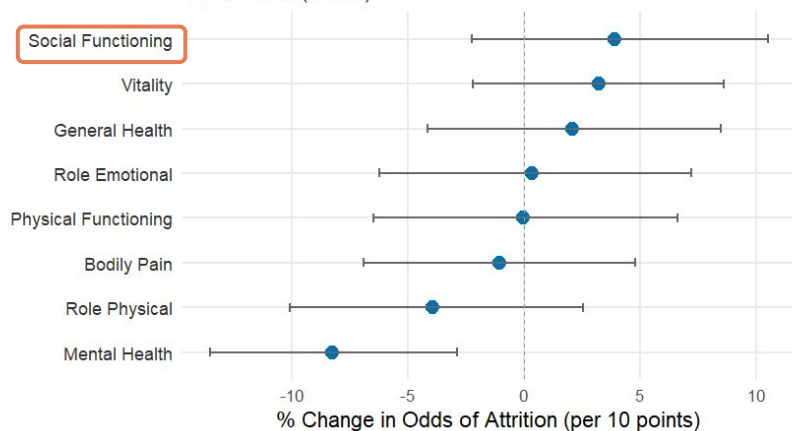
Prediction range: 6.2% - 7.1%



Health Results: Social Functioning

SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Social Functioning

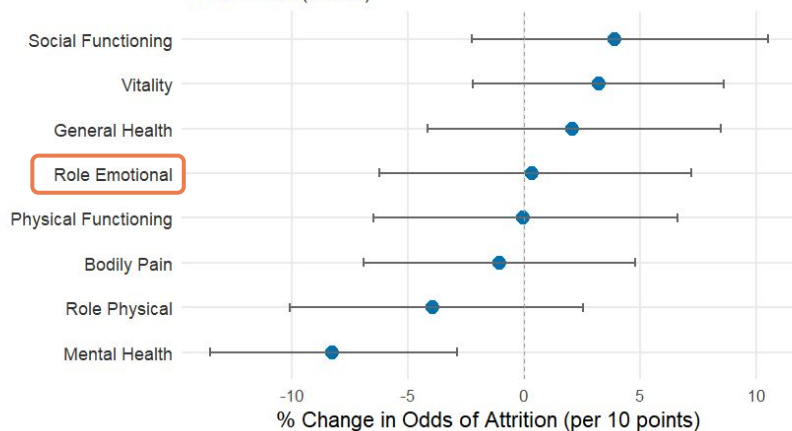
Prediction range: 5.7% - 6.6%



Health Results: Role Emotional

SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Role Emotional

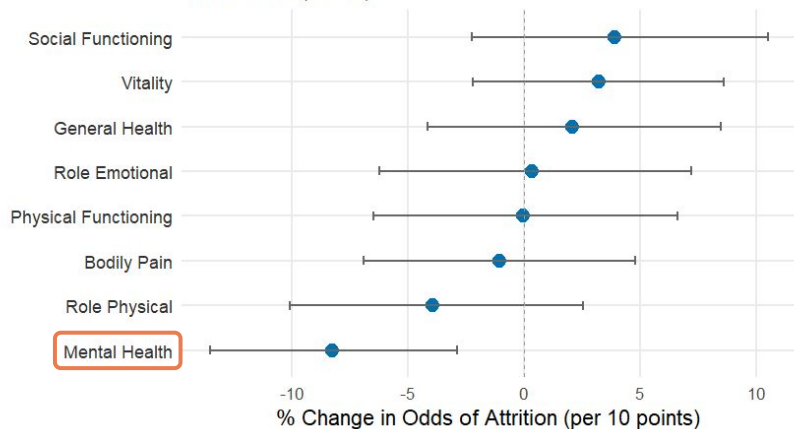
Prediction range: 6.6% - 6.6%



Health Results: Mental Health

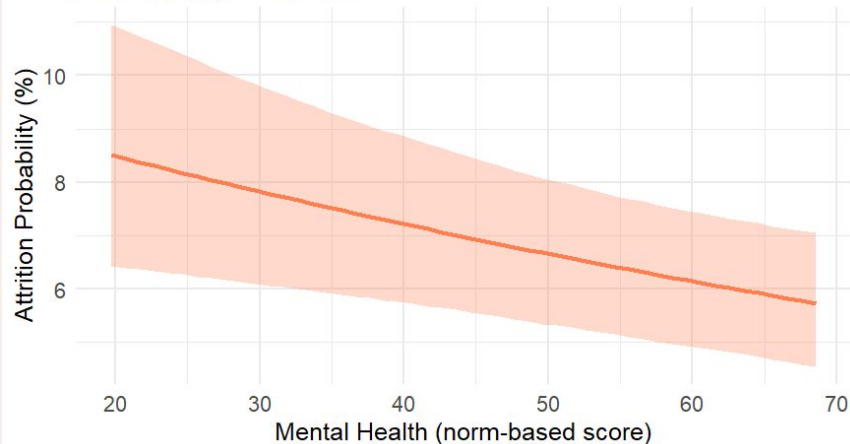
SF-12 Health Subscales: Effect on Attrition

Percentage change in odds per 10-point (1 SD) increase in health score (95% CI)



Predicted Attrition by Mental Health

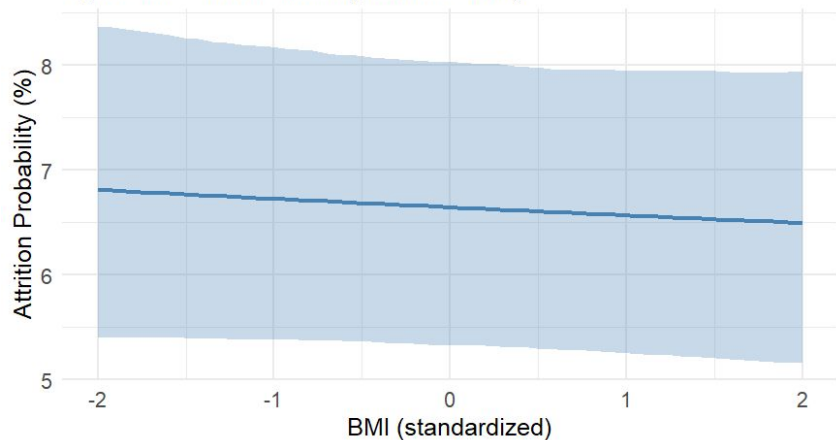
Prediction range: 5.7% - 8.5%



Health Results: BMI and Grip Strength

Predicted Attrition by BMI

Higher BMI → Lower attrition (protective effect)



Odds Change (%): -1.23 °

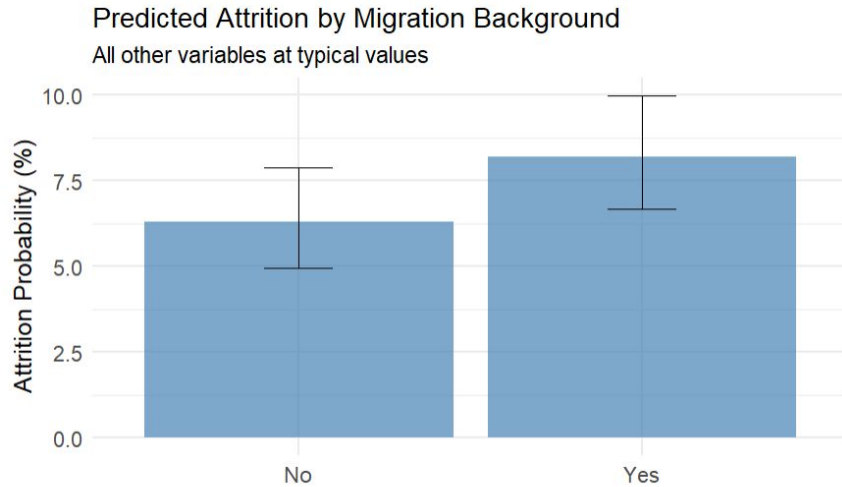
Predicted Attrition by Grip Strength

Higher grip strength → Lower attrition

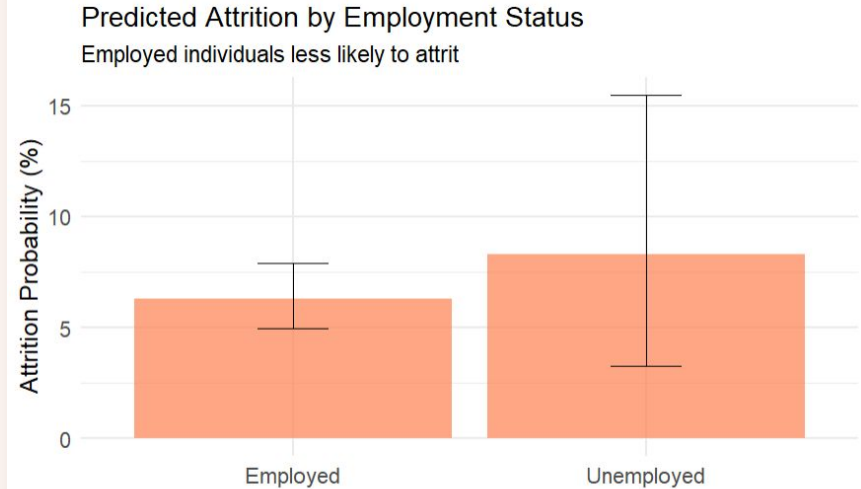


Odds Change (%): -0.20

Control Variable Results: Migration and Employment

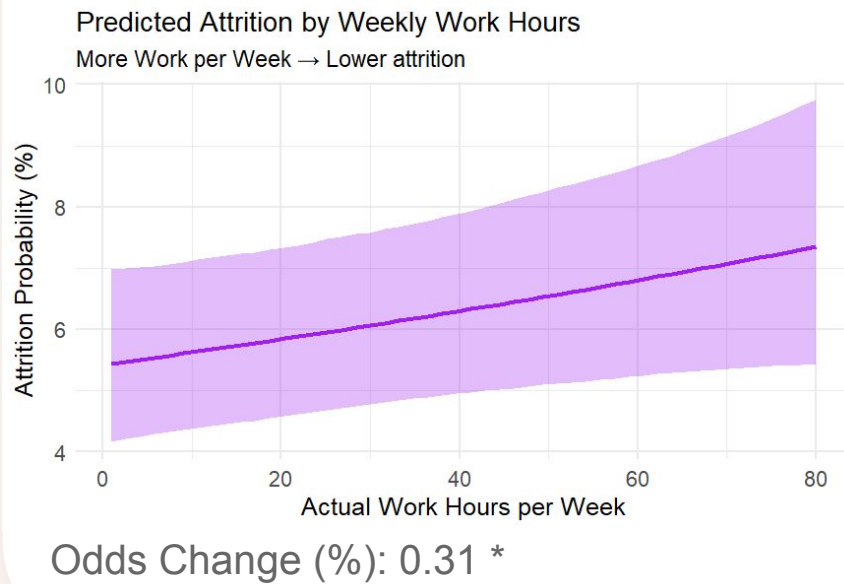
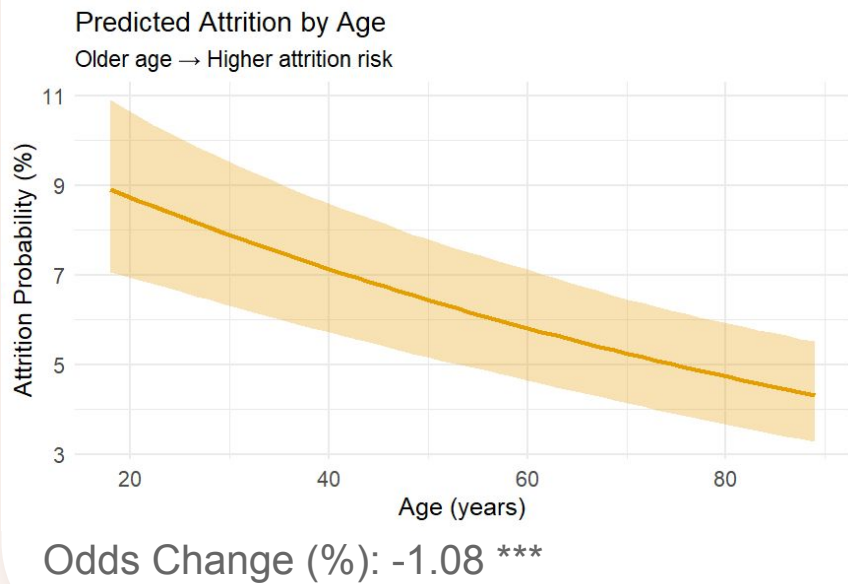


Odds Change (%): 22.23 ***

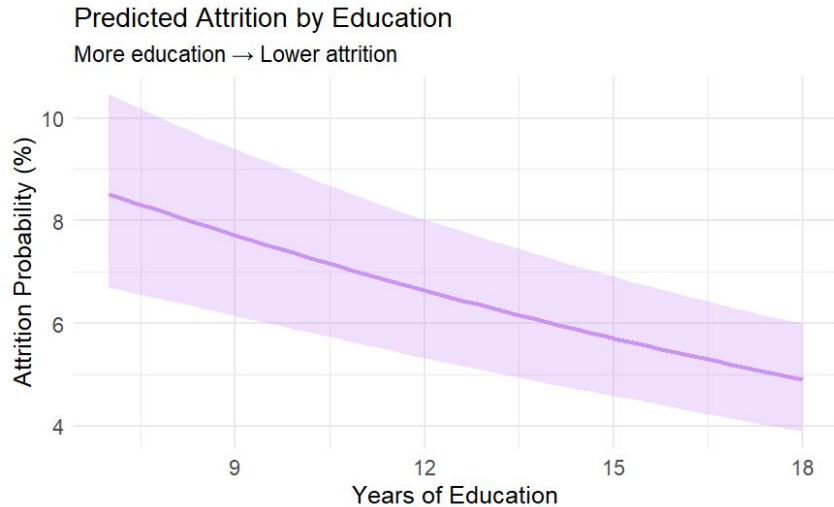


Odds Change (%): -10.05

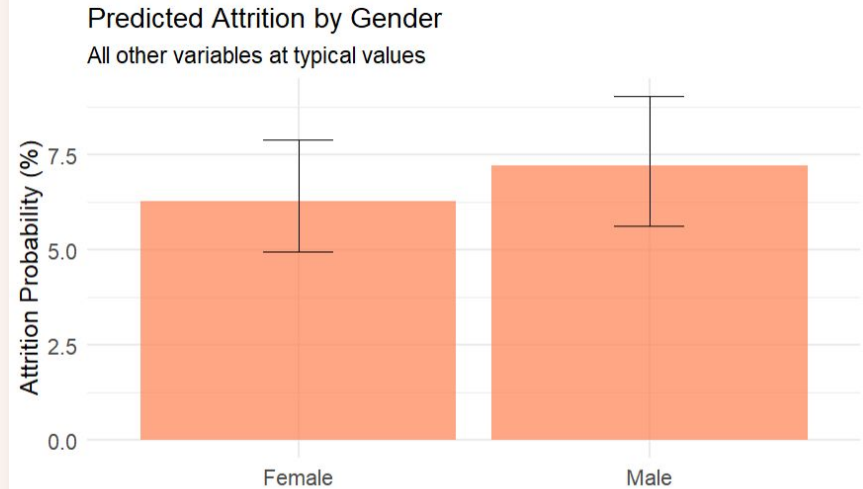
Control Variable Results: Age and Weekly Work Hours



Control Variable Results: Education and Gender



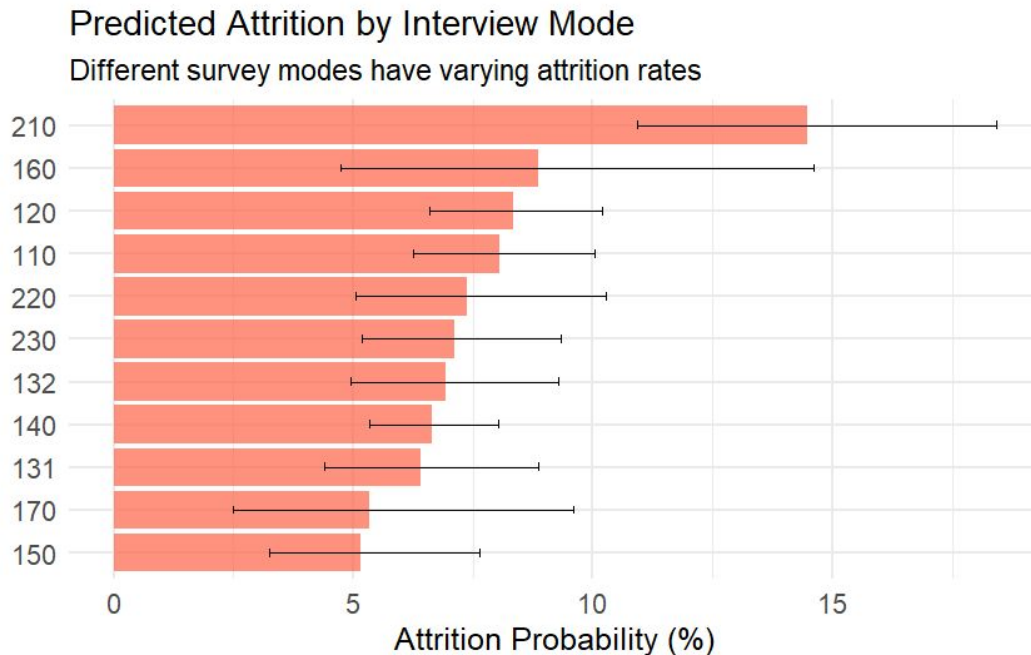
Odds Change (%): -4.08 ***



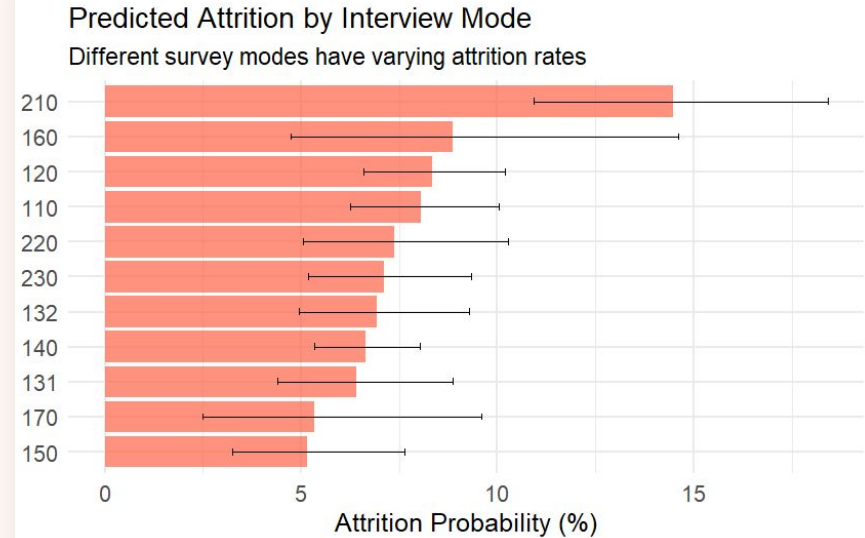
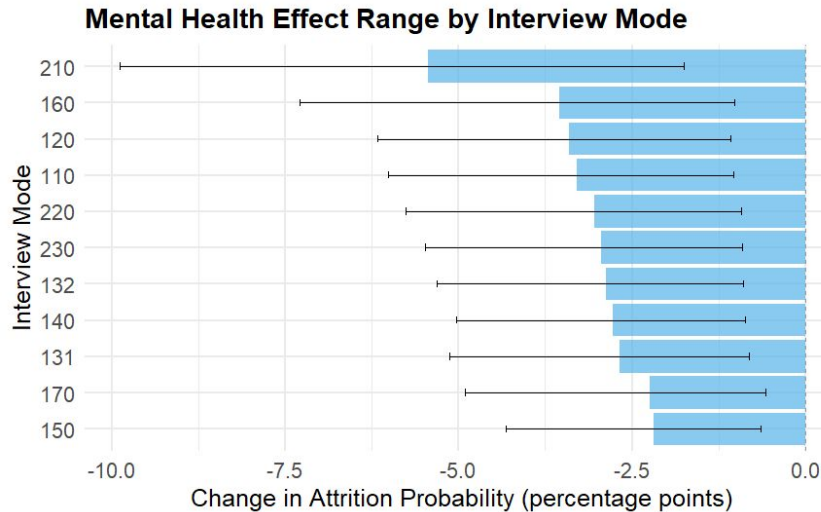
Odds Change (%): -13.30 °

Control Variable Results: Interview Type

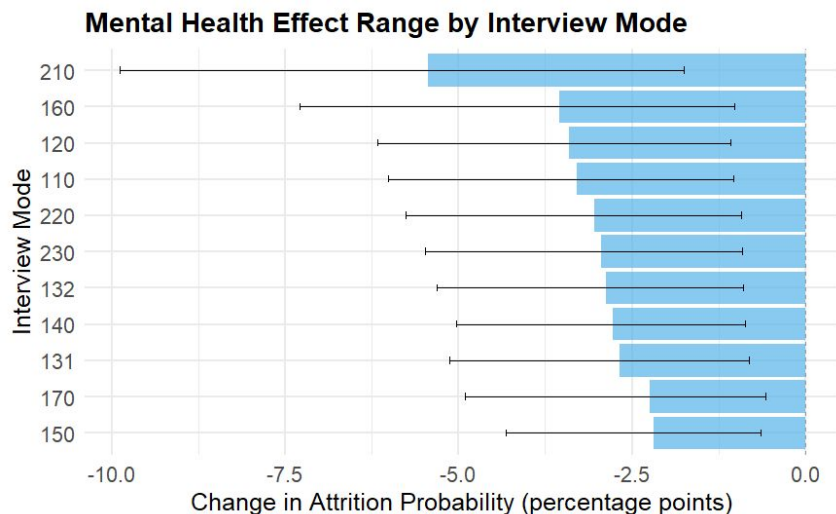
[210] Written, By Mail
[160] CATI since 2021
[120] Written Q. No Interviewer
[110] Oral Interview
[220] Telephone Interview
[230] CAPI by phone
[132] Oral And Written
[140] CAPI - Since 1998 (O)
[131] Written Q. w/ Interviewer
[150] CAWI Since 2014 (BE)
[170] CASI since 2021



Mental Health Effects: Model predictions

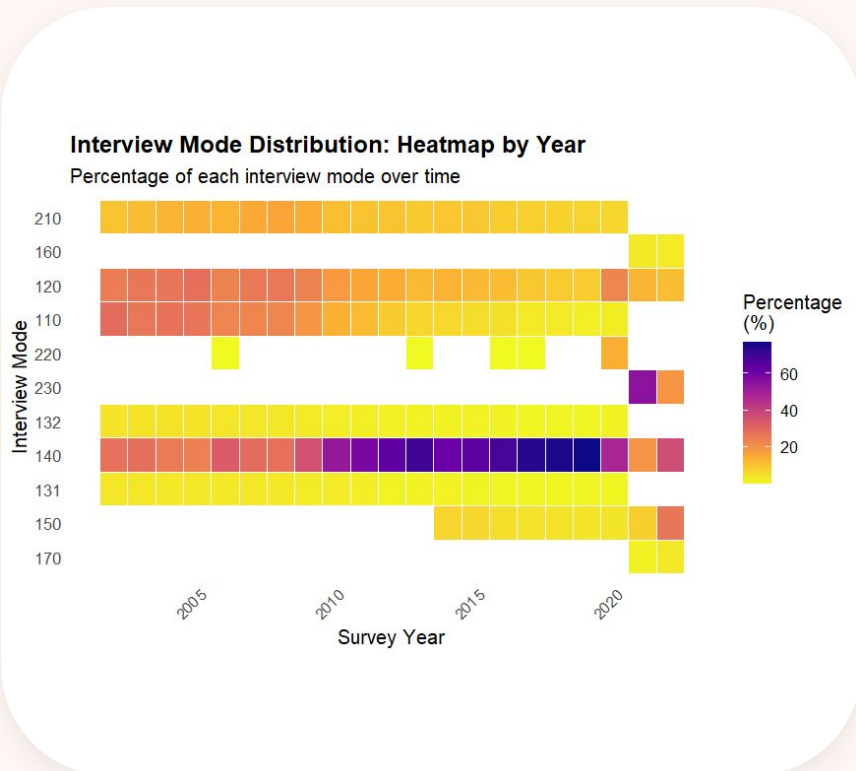


Mental Health Effects: Model predictions



Mode	Method	Interviewer?	Location
210	Mail	No	At home
160	Telephone	Yes	At home
120	Written/Paper	No	At home
110	Face-to-face	Yes	In-person
220	Telephone	Yes	At home
230	Telephone	Yes	At home
132	Oral + Written	Yes	In-person
140	Face-to-face	Yes	In-person
131	Written	Yes	In-person
150	Online/Web	No	At home
170	Computer	Yes	In-person

Mental Health Effects: Model predictions



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131	Written	Yes	In-person
150	Online/Web	No	At home
170	Computer	Yes	In-person

6. Conclusion

To what extent do health factors influence the likelihood of sample attrition in the German Socio-Economic Panel?

What are the implications for the SOEP study design?

6. Conclusion

Health factors' influence on SOEP Attrition

While BMI and Grip Strength are not significant.

Detailed health data shows Mental Health is the main factor predicting attrition

SOEP Study Design Implications

The largest effect of Mental Health is seen for Mail interviews.

Less accountability = More Attrition

→ Highly Correlated with Mental Health

6. Limitations

The results are not Causal.

Note that health data is self reported and subjective.

Health variables are only measured biannually.

→ Physical health may coincide with attrition, which can not be observed.

Thank You for Listening!

